

Symmetric Positive-Definite matrix bounds

Emil Kerimov

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1 Motivating Example

Suppose there are three assets, a, b, c , by knowing the volatility of each of the assets, and the correlation between assets a and b , as well as the correlation between assets b and c , what is the interval of valid correlations between and assets a and c ?

This document presents one general method of addressing this problem, and solves the case of three assets analytically.

2 Definition

Definition 2.1. *A positive-definite matrix is defined as TODO.*

Definition 2.2. *A semi-positive-definite matrix is defined as TODO.*

Definition 2.3. *A principle minor is defined as TODO.*

3 Background Theory

We make use of some known properties of positive-definite (PD) and positive-semi-definite (PSD) matrices.

Theorem 3.1. *A PD matrix must have all positive eigenvalues.*

Proof:

From the definition of PD

TODO

Theorem 3.2. *A PD matrix must have all positive principle minors.*

Proof:

From the definition of PD

TODO

4 Main Idea

All principle majors of a PD matrix, must be positive. Therefore, compute them all, and figure out bounds on the unknown quantities.

4.1 Two Asset

The case $n = 2$ is trivial, due to the definition of correlation being bounded by -1 and 1 .

$$\begin{aligned} |\rho_{a,b}| &\leq 1 \\ \left| \frac{\sigma_{a,b}}{\sigma_a \cdot \sigma_b} \right| &\leq 1 \end{aligned}$$

Since σ_a and σ_b are positive by definition

$$|\sigma_{a,b}| \leq \sigma_a \cdot \sigma_b$$

We show that the positive principle minors method arrives at the same bound.
Let matrix C be a 2x2 matrix,

$$C = \begin{pmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{pmatrix} \quad (1)$$

4.2 3x3

We can

$$C = \begin{pmatrix} 1 & \rho_{ab} & \rho_{ac} \\ \rho_{ab} & 1 & \rho_{bc} \\ \rho_{ac} & \rho_{bc} & 1 \end{pmatrix} \quad (2)$$

Theorem 4.1.

$$|\rho_{ac} - \rho_{ab} \cdot \rho_{bc}| \leq \sqrt{(1 - \rho_{ab}^2) \cdot (1 - \rho_{bc}^2)} \quad (3)$$

Proof:

$$\text{Det}(C) \geq 0$$

TODO

4.3 4x4

We can

$$C = \begin{pmatrix} 1 & \rho_{ab} & \rho_{ac} & x \\ \rho_{ab} & 1 & \rho_{bc} & \rho_{bd} \\ \rho_{ac} & \rho_{bc} & 1 & \rho_{cd} \\ x & \rho_{bd} & \rho_{cd} & 1 \end{pmatrix} \quad (4)$$

Theorem 4.2. Bounds on $x = \rho_{ad}$

$$\begin{aligned} & \left| \rho_{ad} - \frac{(\rho_{a,b} \cdot \rho_{b,d} + \rho_{a,c} \cdot \rho_{c,d}) - \rho_{b,c} \cdot (\rho_{a,b} \cdot \rho_{c,d} + \rho_{a,c} \cdot \rho_{b,d})}{1 - \rho_{b,c}^2} \right| \leq (1 - \rho_{b,c}^2) \sqrt{D} \\ & D = (1 - \rho_{a,b}^2 - \rho_{a,c}^2 - \rho_{b,c}^2 + 2 \cdot \rho_{a,b} \cdot \rho_{a,c} \cdot \rho_{b,c}) \cdot (1 - \rho_{b,c}^2 - \rho_{b,d}^2 - \rho_{c,d}^2 + 2 \cdot \rho_{b,c} \cdot \rho_{b,d} \cdot \rho_{c,d}) \end{aligned}$$

Proof:

$$\begin{aligned}
& \text{Det}(C) \geq 0 \\
& 1 - (\rho_{c,d}^2 + \rho_{b,c}^2 + \rho_{b,d}^2 + \rho_{a,b}^2 + \rho_{a,c}^2) \\
& + 2 \cdot (\rho_{b,c}\rho_{b,d}\rho_{c,d} + \rho_{a,b}\rho_{b,c}\rho_{a,c}) \\
& - 2(\rho_{a,b}\rho_{b,d}\rho_{a,c}\rho_{c,d}) \\
& + (\rho_{a,b}^2\rho_{c,d}^2 + \rho_{a,c}^2\rho_{b,d}^2) \\
& + 2x(\rho_{a,b}\rho_{b,d} + \rho_{a,c}\rho_{c,d}) \\
& - 2x(\rho_{a,b}\rho_{b,c}\rho_{c,d} + \rho_{a,c}\rho_{b,d}\rho_{b,c}) \\
& + x^2(\rho_{b,c}^2 - 1) \geq 0
\end{aligned}$$

This is just a quadratic equation

$$\begin{aligned}
& a = \rho_{b,c}^2 - 1 \\
& b = 2(\rho_{a,b}\rho_{b,d} + \rho_{a,c}\rho_{c,d}) - 2(\rho_{a,b}\rho_{b,c}\rho_{c,d} + \rho_{a,c}\rho_{b,d}\rho_{b,c}) \\
& c = 1 - (\rho_{c,d}^2 + \rho_{b,c}^2 + \rho_{b,d}^2 + \rho_{a,b}^2 + \rho_{a,c}^2) \\
& + 2 \cdot (\rho_{b,c}\rho_{b,d}\rho_{c,d} + \rho_{a,b}\rho_{b,c}\rho_{a,c}) \\
& - 2(\rho_{a,b}\rho_{b,d}\rho_{a,c}\rho_{c,d}) \\
& + (\rho_{a,b}^2\rho_{c,d}^2 + \rho_{a,c}^2\rho_{b,d}^2) \\
& ax^2 + bx + c \geq 0 \\
& |2ax + b| \leq \sqrt{b^2 - 4ac}
\end{aligned}$$

Dividing all sides by 2

$$\begin{aligned}
& a = \rho_{b,c}^2 - 1 \\
& b = (\rho_{a,b}\rho_{b,d} + \rho_{a,c}\rho_{c,d}) - (\rho_{a,b}\rho_{b,c}\rho_{c,d} + \rho_{a,c}\rho_{b,d}\rho_{b,c}) \\
& c = 1 - (\rho_{c,d}^2 + \rho_{b,c}^2 + \rho_{b,d}^2 + \rho_{a,b}^2 + \rho_{a,c}^2) \\
& + 2 \cdot (\rho_{b,c}\rho_{b,d}\rho_{c,d} + \rho_{a,b}\rho_{b,c}\rho_{a,c}) \\
& - 2(\rho_{a,b}\rho_{b,d}\rho_{a,c}\rho_{c,d}) \\
& + (\rho_{a,b}^2\rho_{c,d}^2 + \rho_{a,c}^2\rho_{b,d}^2) \\
& |ax + b| \leq \sqrt{b^2 - ac}
\end{aligned}$$

Simplifying the square root

$$\begin{aligned}
& b^2 - ac = (a,b)^2(b,d)^2 + (a,c)^2(c,d)^2 + (b,c)^2(c,d)^2 + (b,c)^2(b,d)^2 \\
& + (b,c)^2(a,b)^2 + (b,c)^2(a,c)^2 - (a,b)^2(c,d)^2 - (a,c)^2(b,d)^2 \\
& + (2 * (b,c)^2 + (c,d)^2 + (b,d)^2 + (a,b)^2 + (a,c)^2) - 1 \\
& + 4(a,b)(b,c)^2(c,d)(a,c)(b,c) - 2((b,c)(b,d)(c,d) + (a,b)(b,c)(a,c)) - (b,c)^4 \\
& - 2((a,b)^2(b,d)(b,c)(c,d) + (a,b)(b,d)^2(b,c)(a,c) + (a,c)(c,d)^2(a,b)(b,c) \\
& + (a,c)^2(c,d)(b,c)(b,d)) - 2((b,c)^3(b,d)(c,d) + (b,c)^3(a,b)(a,c))
\end{aligned}$$